

Media Metadata Manager - Design Specification

Vision

A desktop-first media metadata manager that treats metadata consistently across photos and videos. The application owns metadata until an explicit Sync operation writes XMP sidecars and/or embeds metadata.

Goals

Fast browsing; unified metadata editing; immutable originals until sync; Adobe-friendly XMP; batch editing; powerful search; proxy generation for smooth video playback.

Core Workflow

Scan folder -> Build catalog -> Edit metadata -> Store pending changes -> Review -> Sync to XMP and/or embedded metadata.

Architecture

Projects: UI (Avalonia), Core, Database, Metadata, Preview, Search, Sync, Plugins.

Tech Stack

C#/F.NET 9, Avalonia, CommunityToolkit.Mvvm, Microsoft DI, SQLite + FTS5, Dapper, ExifTool (stay_open mode), FFmpeg, ImageSharp/SkiaSharp, Serilog.

UI

Folder tree, thumbnail grid, metadata inspector (General/Camera/Video/XMP/History), preview panel with image zoom and low-resolution video playback, status bar.

Metadata

Virtual metadata layer with pending edits. Source of truth stored locally. Sync writes XMP sidecars and/or embeds metadata depending on format and user settings.

Database

Tables: Media, Metadata, Keywords, PendingChanges, PreviewCache, SearchIndex.

Sync

Conflict detection, preview of changes, backup originals option, preserve timestamps where possible, resumable operations.

Plugins

Interfaces: IMetadataProvider, IPreviewProvider, ISyncProvider, IMediaAnalyzer.

Future

AI analyzers, duplicate detection, semantic search, Resolve/Premiere exporters, cloud sync.